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Are Consumers Willing to Disclose Information to AI-Based Chatbot? The Roles of Anthropomorphism and Regulatory Focus

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ABSTRACT

Though anthropomorphism has become a prevalent design strategy in artificial intelligence, scant empirical evidence has examined whether anthropomorphizing recommendation chatbots affects consumers' willingness to disclose personal information. This study therefore explores how chatbot anthropomorphism influences consumers' information-disclosure intention and the psychological mechanisms underlying this relationship. Findings from three experimental studies indicate that heightened chatbot anthropomorphism enhances consumers' information-disclosure intention (Studies 1, 2a, and 2b). This effect is mediated by perceived benefits (Study 1). Importantly, regulatory focus moderates the effect of chatbot anthropomorphism on information-disclosure intention (Studies 2a and 2b): promotion-focused consumers are more willing to disclose personal information when the chatbot anthropomorphism is high (vs. low), whereas prevention-focused consumers remain unaffected by varying levels of anthropomorphism. These findings shed light on how to design effective chatbot interfaces and how to target the right consumers.

1 | Introduction

Amid the ongoing intelligent revolution, artificial intelligence (AI) technologies empower marketers and managers with viable options to deliver superior services, enhance consumer engagement, and gain competitive advantages. Not surprisingly, AI-enabled recommendation systems are increasingly pervasive in the online marketplace. Alibaba and JD (http://jd.com), for instance, apply AI and deep learning to monitoring consumers' actions and generating product recommendations, while beauty brands like Curology and Proven deploy AI to understand their consumers and administer recommendations about makeup and skincare. In this trend, AI-based chatbots, as autonomous agents supporting communication and exchange with human users, are increasingly used to communicate with consumers, make recommendations, simplify online searches, and ease the purchasing processes (Kim et al. 2023; Luo et al. 2019; Schanke et al. 2021).

To deliver real-time, highly customized conversations and recommendations, these chatbots require consumers to disclose personal information such as demographic details, location, interests, and activities (Al-Natour et al. 2021; Bol et al. 2018; Schanke et al. 2021). Such data constitute a strategic asset that enables firms to discern consumer preferences and craft tailored offerings (Hayes et al. 2021; Johnson et al. 2020). In specialized or complex decision contexts, the need for granular personal data intensifies to ensure individualized service delivery (Xu

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et al. 2017, 2011). Without early consumer disclosure to AI assistants, the value realization and widespread adoption of AI technologies may be significantly hindered (Al-Natour et al. 2021). It is therefore imperative to identify the factors that influence consumers' willingness to disclose personal information to AI-based chatbots.

As AI technologies advance, firms increasingly deploy anthropomorphism—the attribution of human-like traits and characteristics to non-human entities—as a promising instrument for customer-service interactions and marketing communications. For instance, Taobao's AI assistant "Xiaomi" greets users with a human-like avatar and first-person dialog: "Hi there! I'm Xiaomi, happy to help you today!" Similarly, Fidelity has launched a highly anthropomorphic financial-advisor product named "Cora," allowing clients to converse with the virtual advisor as naturally as they would with a human expert. When an AI assistant is perceived as anthropomorphic, it functions as a social influencer capable of shaping consumer attitudes and behaviors (Munnukka et al. 2022). Anthropomorphic agents are typically evaluated as more intelligent, competent, and empathetic (Cheng 2022; Kim et al. 2022; Li and Wang 2022). A strand of research has revealed the positive downstream effects of AI anthropomorphism on consumers' perceptions (e.g., social presence, trust, perceived usefulness, and performance expectation), behavioral intentions (e.g., purchase intention and usage intention), and transaction outcomes (e.g., conversion rate) (Konya-Baumbach et al. 2023; Li and Wang 2022; Munnukka et al. 2022; Schanke et al. 2021; Yang et al. 2021). Yet, it remains not clearly characterized whether AI anthropomorphism exerts an influence on consumers' willingness to disclose personal information. Hence, the present study seeks to address this gap.

Drawing on the privacy calculus theory (Dinev and Hart 2006; Duan and Deng 2022; Kehr et al. 2015), we propose that chatbot anthropomorphism positively influences consumers' information-disclosure intention. Privacy calculus theory posits that information-disclosure decisions reflect a rational trade-off between perceived benefits and perceived risks. The fact that anthropomorphism positively affects perceived competence, perceived usefulness, and performance expectation implies that anthropomorphism might have a positive impact on perceived benefits (Li and Wang 2022; Yang et al. 2021; Zhang et al. 2021). Meanwhile, some studies indicate that anthropomorphism can alleviate perceived risks by increasing perceived control (Chan and Gohary 2022; Kim and McGill 2011). Accordingly, we theorize that chatbot anthropomorphism influences information-disclosure intention both by enhancing perceived benefits and diminishing perceived risks.

Beyond verifying this core proposition, the study further seeks to delineate the boundary conditions that may amplify or diminish the effect of chatbot anthropomorphism. Its primary objective is to provide a holistic account of the conditions under which chatbot anthropomorphism is most effective. Guided by regulatory focus theory, this study specifically investigates whether regulatory focus serves as a critical boundary condition that shapes how chatbot anthropomorphism influences consumers' information-disclosure intention. Regulatory focus theory is a framework that explains the relationship between motivations and actions for goal pursuits (Cesario et al. 2008; Higgins et al. 1997). It posits that promotion-focused individuals are attentive to accomplishments, gains, and benefits because they pursue their ideal selves, whereas prevention-focused individuals are attentive to pains, losses, and risks because they pursue their aught selves (Higgins et al. 1997; Kim et al. 2023; Wang et al. 2020). Thus, promotion-focused consumers would be more likely to seek the benefits while neglecting the risks involved in information disclosure and thus would be more inclined to disclose personal information when the chatbot is highly (vs. less) anthropomorphized. By contrast, prevention-focused consumers would be risk-avoidant and attach less importance to the benefits of chatbot anthropomorphism.

Taken together, to address the gap in the literature, this study answers the following questions:

Q1. Does chatbot anthropomorphism affect consumers' information-disclosure intention?

Q2. What mechanisms underlie the association between chatbot anthropomorphism and information-disclosure intention?

Q3. Which boundary condition moderates this relationship? Specifically, does regulatory focus serve as a boundary condition?

The rest of the paper is structured as follows. We first review related literature and provide the rationale for hypotheses development. Afterward, research methodologies of and results from three experimental studies are reported to test the hypotheses. Finally, we summarize the main research findings, discuss the theoretical contributions and practical implications, and point out directions for future research.

2 | Theoretical Background and Research Hypotheses

2.1 | Information Disclosure and Privacy Calculus Theory

Given its pivotal role in the digital economy, personal information is often characterized as the "new oil" (Spiekermann and Novotny 2015). Consequently, researchers have strived to uncover the antecedents of personal information disclosure. Established determinants include personal innovativeness (Xu et al. 2011), privacy statements (Schwaig et al. 2006; Zhang et al. 2020), privacy protection (Schomakers et al. 2022; Zhang et al. 2020), institutional trust (Kehr et al. 2015), consumer-brand relationship (Hayes et al. 2021), perceived benefits (Schomakers et al. 2022; Xu et al. 2011), and perceived risks (Hayes et al. 2021; Xu et al. 2011).

Various theories have been proposed to understand individuals' privacy behavior, including privacy calculus theory (Dinev and Hart 2006), utility maximization theory (Awad and Krishnan 2006), and social exchange theory (Cook and Rice 2006). After reviewing 15 theories that involve privacy-driven behavior, Li (2012) suggested that privacy calculus theory is a common thread connecting different theories for

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examining consumer information disclosure. Privacy calculus theory posits that individuals decide whether to disclose personal data by weighing anticipated benefits (e.g., monetary rewards, personalized pricing, customized offerings) against anticipated privacy losses (e.g., unauthorized access or misuse of data) (Dinev and Hart 2006; Duan and Deng 2022; Kehr et al. 2015). When perceived benefits outweigh perceived risks, consumers are motivated to share their personal information. Conversely, consumers refrain from sharing personal information (Kehr et al. 2015; Schumacher et al. 2023; Xu et al. 2011).

Privacy calculus theory has been successfully implemented for explaining information disclosure behavior and new technology adoption in a variety of settings, including personalized advertising (Hayes et al. 2021; Xu et al. 2011), location-based advertising (Ryu 2023), social networks (Trepte et al. 2017), and smart technology (Schomakers et al. 2022). For instance, Xu et al. (2011) employed privacy calculus theory to explore consumers' willingness to have personal information used in location-aware marketing, revealing that perceived benefits positively affect and perceived risks negatively affect the perceived value of information disclosure, which in turn positively affects willingness to disclose personal information. Additionally, Hayes et al. (2021) demonstrated that both perceived benefits and perceived risks influence consumers' willingness to have personal information used in ad personalization in social media, with brand relationship strength moderating these effects.

2.2 | AI-Based Agents and the Role of Anthropomorphism

AI technologies have been deployed across a wide spectrum of consumer domains, including smart voice assistants (Guha et al. 2023; Mishra et al. 2022) and recommendation services (Garvey et al. 2022; Longoni and Cian 2020; Lv and Huang 2022). A Salesforce survey indicates that AI-driven recommendation is likely to become a widely preferred technology by online marketers (Columbus 2019). The switch from using human employees to using AI agents in administering product and service offers is advantageous for businesses given the superior information-processing abilities and reduced labor costs brought about by AI technologies (Kumar et al. 2016; Luo et al. 2019).

To foster effective human–AI interaction and optimize agent performance, social factors such as anthropomorphism have gained notable attention (Schanke et al. 2021; Yang et al. 2021; Zhang et al. 2021). Anthropomorphism is considered inevitable and should be taken into consideration when developing human–computer interfaces (Caporael 1986; Schanke et al. 2021). Anthropomorphism, defined as the attribution of human-like traits and characteristics to non-human entities like animals and machines, primarily exists because humans attempt to make sense of the behavior or action of non-human entities in a way that is easy to understand (Epley et al. 2007). AI agents typically exhibit anthropomorphism along two dimensions: physical characteristics (e.g., human-like appearance and behavior) and psychological capabilities (e.g., empathy, humor, and emotional experience) (Kim et al. 2022; Xie and Lei 2022). Encountering such cues activates human schemas that guide subsequent evaluations and behaviors (Qiu and Benbasat 2009; Yang et al. 2021).

Since AI technologies are widely used to replace human employees and serve as crucial customer touchpoints, the anthropomorphism of AI technologies exerts a profound influence on consumer responses (Schanke et al. 2021; Xie and Lei 2022; Zhang et al. 2021). Compared to less anthropomorphized AI-based agents (e.g., service assistant, product recommender, and text-based chatbot), highly anthropomorphized ones trigger favorable downstream psychological and behavioral outcomes, including heightened social presence (Munnukka et al. 2022), trust (Konya-Baumbach et al. 2023; Mulcahy et al. 2024; Qiu and Benbasat 2009), quality perception (Bhatnagr et al. 2024; Munnukka et al. 2022), perceived intelligence (Malhotra and Ramalingam 2023), performance expectation (Yang et al. 2021; Zhang et al. 2021), perceived usefulness (Li and Wang 2022; Zhang and Wang 2023), usage intention (Qiu and Benbasat 2009), and purchase intention (Jin and Youn 2023; Konya-Baumbach et al. 2023).

By operationalizing chatbot anthropomorphism through a set of three anthropomorphic features (e.g., humor, social presence, and communication delays), Schanke et al. (2021) observed that chatbot anthropomorphism increases transaction conversion. Based on some descriptive evidence, Schanke et al. (2021) conjectured that the positive impact of anthropomorphism on transaction outcomes may partly stem from consumers' increased intention to share personal information that is essential for completing the transaction. As illustrated in Table 1, a small but growing body of work has examined the intersection of anthropomorphism and privacy (Aw et al. 2024; Pizzi et al. 2023; Uysal et al. 2022; Xie and Lei 2022). For instance, Aw et al. (2024) suggest that the perceived anthropomorphism of robo-advisors has a positive impact on perceived justice, which further reduces privacy concerns and perceived intrusiveness. Contrarily, it is revealed by Uysal et al. (2022) that the anthropomorphism of AI assistants creates data privacy concerns due to increased human identity threat and disempowerment. A recent study in the context of AI-driven public services demonstrates that individuals' privacy concerns are neither influenced by the quantity of personal data they are required to disclose nor by anthropomorphic interfaces (Willems et al. 2023).

It can be seen that prior research has extensively examined the downstream psychological and behavioral outcomes of anthropomorphic AI-based agents, but it remains underexplored whether and how anthropomorphic AI-based agents affect consumers' information-disclosure intention. Though a few studies have attempted to probe the association between anthropomorphism and privacy-related issues, there are still some notable gaps in the literature. First, most of them have not examined privacy disclosure as a focal outcome in human–AI interaction (Aw et al. 2024; Uysal et al. 2022; Xie and Lei 2022). Second, they have reported mixed findings regarding the anthropomorphism—privacy concern association (Aw et al. 2024; Uysal et al. 2022; Willems et al. 2023; Xie and Lei 2022). Third, the psychological mechanisms (particularly positive psychological processes like perceived benefits) underpinning and the boundary condition of the impact

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TABLE 1 | Relevant literature on AI anthropomorphism and privacy-related issues.

Study	Context	Independent variables	Dependent variables	Psychological mechanism	Moderator	Main insights
Xie and Lei (2022)	Service robot	Service robot anthropomorphism (SRA)	Usage intention	Privacy concerns	Information sensitivity, service personalization	SRA nonlinearly affects privacy concerns and usage intention. The negative impacts of SRA are attenuated when information sensitivity is low and service personalization is high.
Uysal et al. (2022)	AI assistants	Anthropomorphism	Consumer well-being, privacy concerns, satisfaction	Human identity threat, trust, empowerment	Relationship closeness	Anthropomorphism threatens users' identity, disempowers them, creates data privacy concerns and ultimately undermines their well-being.
Pizzi et al. (2023)	Customer-service chatbot	Anthropomorphism, gaze direction	Willingness to disclose, future intentions	Warmth perceptions, competence perceptions, skepticism, trust	Theory of mind	Competence, skepticism, and trust fully mediated the relationship between anthropomorphism and consumers' disclosure/future intentions.
Willems et al. (2023)	AI-driven public service	Information to share, anthropomorphism, privacy concerns	Willingness to download	/	/	Citizens' willingness to download an AI-driven public service mobile app is not affected by the anthropomorphic interface and the amount of personal information required to share.
Aw et al. (2024)	Robo-advisor	Perceived anthropomorphism, perceived autonomy	Resistance to robo-advisors	Perceived justice, privacy concern, perceived intrusiveness	/	Perceived anthropomorphism and perceived autonomy positively influence perceived justice and further negatively impacts privacy concerns and perceived intrusiveness, which positively influence resistance to robo-advisors.
Lefkeli et al. (2024)	Customer-service agent	Service agent (AI vs. human)	Brand trust	Sense of exploitation	Anthropomorphism, information confidentiality	Consumers exhibit lower trust in the brand when disclosing information to AI (vs. humans). This effect is mitigated when the confidentiality of the information is safeguarded or when AI is anthropomorphized.

(Continues)

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TABLE 1 | (Continued)

Study	Context	Independent variables	Dependent variables	Psychological mechanism	Moderator	Main insights
This study	Product recommendation chatbot	Chatbot anthropomorphism	Information-disclosure intention	Perceived benefit, perceived risk	Regulatory focus	Chatbot anthropomorphism positively influences consumers' information-disclosure intention through perceived benefit. Promotion-focused consumers exhibit higher disclosure intention when chatbot anthropomorphism is high (vs. low), while prevention-focused consumers are indifferent to chatbot anthropomorphism.

of anthropomorphism on consumers' information-disclosure intention remain underspecified. Given that personal disclosure (e.g., demographics, location, interests) is a prerequisite for personalized AI recommendations (Al-Natour et al. 2021; Bol et al. 2018; Kim et al. 2022), a deeper investigation is warranted.

Linking the privacy calculus theory to anthropomorphism research, we theorize that chatbot anthropomorphism has a positive impact on consumers' information disclosure intention due to enhanced perceived benefits and reduced perceived risks. On the one hand, prior studies demonstrate that the human-like features of service agents evoke social responses (e.g., emotional attachment, perceived affinity, and trust) from consumers, which encourage interactions between consumers and agents and result in higher perception of utility (Belanche et al. 2021; Duffy 2003). As previously discussed, the perceived anthropomorphism of AI-based agents positively influences performance expectation, perceived usefulness, and service quality, which are regarded as benefits that consumers could acquire from using such AI technologies (Al-Natour et al. 2021; Belanche et al. 2021; Li and Wang 2022; Yang et al. 2021; Zhang et al. 2021). Hence, chatbot anthropomorphism is highly likely to increase the perceived benefits (e.g., getting customized and satisfactory recommendations) of using the chatbot. On the other hand, existing literature also shows that compared to a less anthropomorphized entity, a highly anthropomorphized entity allows consumers to apply social expectations to it, perceive greater control over it, and ultimately perceive fewer risks (Chan and Gohary 2022; Kim and McGill 2011). Accordingly, we surmise that chatbot anthropomorphism is likely to alleviate the perceived risks associated with disclosing personal information. Formally, we have:

H1. *Chatbot anthropomorphism has a positive effect on consumers' information-disclosure intention.*

H2a. *Perceived benefits mediate the effect of chatbot anthropomorphism on consumers' information-disclosure intention. Specifically, higher anthropomorphism increases perceived benefits, which in turn enhance information-disclosure intention.*

H2b. *Perceived risks mediate the effect of chatbot anthropomorphism on consumers' information-disclosure intention. Specifically, higher anthropomorphism reduces perceived risks, thereby enhancing information-disclosure intention.*

2.3 | Regulatory Focus Theory

Regulatory focus theory provides a framework to explain the relationship between motivations and goal pursuit (Cesario et al. 2008; Higgins et al. 1997). According to the theory, individuals who are predominantly promotion-focused are motivated to pursue their ideal selves through accomplishment and success, while individuals who are predominantly prevention-focused are motivated to attain ought selves through undertaking responsibilities and seeking safety (Aaker and Lee 2001; Higgins et al. 1997). Different strategies are employed by the two types of individuals to attain their goals. Promotion-focused

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individuals are inclined to approach novel opportunities, satisfy their desires, and maximize gains and benefits. In contrast, prevention-focused individuals, who are sensitive to negative outcomes and vigilant in minimizing risk, loss, and pain, tend to adopt conventional solutions, maintain the status quo, and avoid making mistakes (Baek and Kim 2023; Kim et al. 2023; Wang et al. 2020).

Consumer research has widely examined the role of regulatory focus in decision-making. In the context of celebrity endorsement for international destination marketing, promotion-focused consumers prefer local (vs. source) celebrity endorsement because the celebrity-tourist incongruity perception increases the novelty of future travel, while prevention-focused consumers prefer source (vs. local) celebrity endorsement because the celebrity-tourist congruity perception reduces the uncertainty of future travel (Li et al. 2022). Likewise, Kim et al. (2023) reported that promotion-focused consumers exhibit higher purchase intention when chatbot advertising personalization is high (vs. low) because they are attentive to the benefits derived from information disclosure, whereas prevention-focused consumers display the opposite pattern due to their heightened sensitivity to the associated risks.

Extending this logic to recommendation chatbots, we expect that regulatory focus moderates the relationship between chatbot anthropomorphism and consumers' information-disclosure intention. Privacy calculus theory conceptualizes information disclosure as a rational trade-off between perceived benefits and perceived risks (Al-Natour et al. 2021; Xu et al. 2011). Building on this theory, chatbot anthropomorphism is likely to affect information disclosure by altering benefit and risk perceptions. Regulatory focus theory complements this account by specifying how individuals differentially weigh benefits against risks. Promotion-focused consumers are more likely to seek benefits and downplay risks associated with information disclosure (Kim et al. 2023), and thus would be more apt to disclose personal information when the chatbot is highly (vs. less) anthropomorphized. In contrast, prevention-focused consumers value potential risks and are less likely to attend to the potential benefits gained from chatbot anthropomorphism (Kim et al. 2023; Wang et al. 2020). Consequently, prevention-focused consumers would be largely indifferent to variations in chatbot anthropomorphism and show equivalent levels of perceived benefits, perceived risks, and information-disclosure intention. Thus, we propose that:

H3. *Regulatory focus moderates the effect of chatbot anthropomorphism on consumers' information-disclosure intention. Specifically, chatbot anthropomorphism enhances the information-disclosure intention of promotion-focused consumers but exerts no significant influence on that of prevention-focused consumers.*

Figure 1 presents the conceptual model of the study.

3 | Study 1

The objectives of Study 1 were twofold: First, we examined the basic role of chatbot anthropomorphism in affecting consumers' information-disclosure intention. Second, we tested if perceived benefits and perceived risks served as mediators in this relationship.

3.1 | Design and Participants

A single factor (anthropomorphism: high, low) between-subjects design was employed in this study. One hundred seventy-one consumers (49.1% female; $M_{age} = 20.6$ years, SD = 2.66) from a local university participated in the online experiment in exchange for small compensation. The specific demographic information of the participants can be found in Appendix F in Supporting Information. The experiment was implemented via Wenjuanxing (http://wjx.com/), a leading survey platform in China. The participants were randomly assigned to different conditions in the experiment.

3.2 | Stimuli and Procedure

A fictitious text-based chatbot interface was devised to eliminate any potential confounding effect of extant knowledge about real chatbots. Participants were instructed to imagine that an e-commerce site they had previously used had launched an AI recommendation system and invited them to try it. Afterward, the participants were shown a chatbot interaction interface. Following approaches used by prior research (Jin and Youn 2023; Kim et al. 2023; Puzakova and Kwak 2017), chatbot anthropomorphism was manipulated by chatbot profile photo, name, and first or third-person description. To be specific, in the high anthropomorphism condition, the interface depicted the chatbot with a photo of a female recommendation agent and a human name (i.e., "Xiao Ai"), and the speech was presented in the first person (e.g., "The more comprehensive information you provide me, the more accurately I can offer personalized product recommendation services to you."). Conversely, in the low anthropomorphism condition, the interface depicted the chatbot with a photo without human appearance features and the speech was presented in the third person (e.g., "The more comprehensive information you provide me, the more accurately the system can offer personalized product recommendation services to you."). Apart from these manipulations, other elements of the interfaces across the two conditions remained identical (see Appendix A in Supporting Information).

A pretest ($n = 75$) was performed to confirm if the manipulations were appropriate. Participants of the pretest were asked to evaluate the perceived anthropomorphism ("The AI application feels like a person," and "The AI application has a mind of its own" from Waytz et al. [2010]; inter-item correlation: $r = 0.682$, $p < 0.001$), positive affect (I feel "interested/excited/active;" $\alpha = 0.851$), and negative affect (I feel "upset/irritable/hostile" from Watson et al. [1988]; $\alpha = 0.874$) after viewing the stimuli. Results showed that the chatbot in the high anthropomorphism condition elicited a higher level of perceived anthropomorphism than that in the low anthropomorphism condition ($M_{high\text{-}anthr} = 5.76$, $M_{low\text{-}anthr} = 4.26$; $F(1, 73) = 25.330$, $p < 0.001$). However, positive and negative affect did not differ between the high and low anthropomorphism conditions (positive affect:

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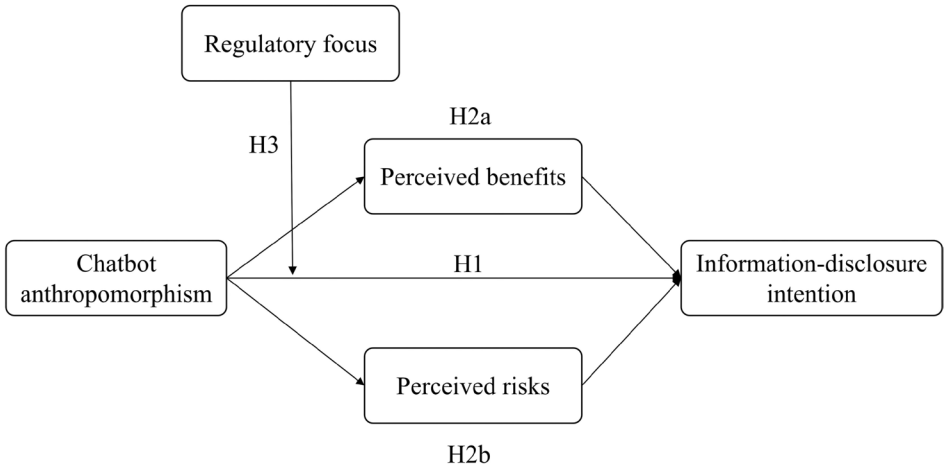


FIGURE 1 | The conceptual framework of the study.

$M_{\text{high-anthr}} = 3.59$, $M_{\text{low-anthr}} = 3.42$; $F(1, 73) = 0.562$, $p = 0.456$; negative affect: $M_{\text{high-anthr}} = 1.31$, $M_{\text{low-anthr}} = 1.50$; $F(1, 73) = 1.185$, $p = 0.280$).

In the main experiment, after reading the materials, participants were invited to complete a battery of measures on their information-disclosure intention ($\alpha = 0.908$), perceived benefits ($\alpha = 0.751$), and perceived risks ($\alpha = 0.903$). The information-disclosure intention was measured by three items adapted from Wang et al. (2016) (see Appendix E in Supporting Information). Three items derived from Dinev et al. (2013) were used to measure perceived benefits. Four items were adapted from Dinev et al. (2013) to assess perceived risks. All items were assessed on 7-point Likert scales ranging from 1 (strongly disagree) to 7 (totally agree). After that, participants provided demographic information.

It should be noted that all experimental materials (e.g., chatbot interaction interfaces, scenario descriptions, task instructions) and questionnaires in the present research (including Studies 1, 2a, and 2b) were developed in Mandarin Chinese, the official and most widely used language in mainland China, which aligns with the cultural and linguistic background of our participants. To ensure the validity of the scales adapted from English-language literature, these scales underwent rigorous back-translation (Brislin 1970): two bilingual researchers translated them into Mandarin, a third bilingual researcher back-translated the Mandarin version into English, and any discrepancies identified were resolved via team discussion.

3.3 | Results

3.3.1 | Main Effect of Anthropomorphism

One-way (anthropomorphism: high, low) ANOVAs were conducted for information-disclosure intention, perceived benefits, and perceived risks. As illustrated in Figure 2, participants reported higher information-disclosure intention ($M_{\text{high-anthr}} = 4.52$, $M_{\text{low-anthr}} = 3.99$; $F(1, 169) = 6.719$, $p = 0.010$) and perceived benefits ($M_{\text{high-anthr}} = 5.13$, $M_{\text{low-anthr}} = 4.78$; $F(1, 169) = 6.128$, $p = 0.014$) in the high (vs. low) anthropomorphism condition, supporting H1. However, the effect of anthropomorphism on perceived risks was not significant ($M_{\text{high-anthr}} = 5.60$, $M_{\text{low-anthr}} = 5.55$; $F(1, 169) = 0.129$, $p = 0.720$).

3.3.2 | Mediation Effects

The potential mediating roles of perceived benefits and perceived risks were investigated through a bootstrapping analysis using model 4 in PROCESS macro (5000 resamples; see Hayes 2013). Perceived benefits and perceived risks were entered into the model as joint mediators. The results demonstrated that the effect of chatbot anthropomorphism on information-disclosure intention was mediated by perceived benefits ($b = 0.335$, 95% CI [0.0746, 0.6141]) but not by perceived risks ($b = -0.009$, 95% CI [−0.0695, 0.0532]). Consequently, H2a was supported; H2b was not.

3.4 | Discussion

In line with our expectation, the findings of Study 1 provided evidence that consumers' information-disclosure intention varied with chatbot anthropomorphism, as a chatbot with high (vs. low) anthropomorphism induced higher privacy disclosure intention. In addition, the mediating effect of perceived benefits in the relationship between chatbot anthropomorphism and consumers' information-disclosure intention was confirmed. Perceived risks did not play any mediating role.

4 | Study 2

The purpose of Study 2 was to test if regulatory focus moderated the effect of chatbot anthropomorphism on consumers' information-disclosure intention. According to the extant literature, regulatory focus can be a chronic personal trait or can be situationally evoked by message framing or certain tasks (Kim et al. 2023; Li et al. 2022). Following approaches used in prior research, two studies were conducted to examine the moderating role of regulatory focus, as Study 2a measured regulatory focus as a personal trait and Study 2b manipulated regulatory focus as a situational variable. Perceived benefits and perceived risks were also evaluated in both studies to

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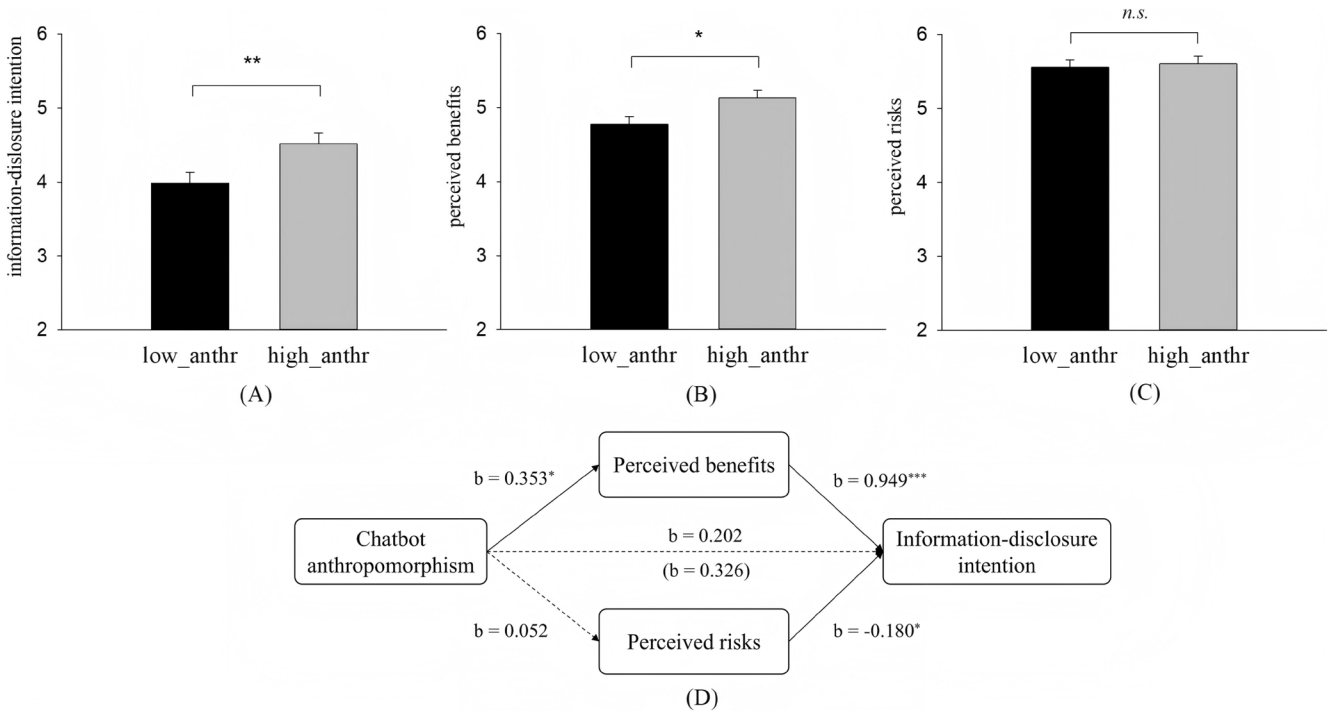


FIGURE 2 | Results of Study 1. (A) Information-disclosure intention; (B) perceived benefits; (C) perceived risks. The error bars denote standard error of the mean; (D) Mediation results. The value in parentheses indicates the aggregated mediation effect. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

test if they mediated the interaction effect of chatbot anthropomorphism and regulatory focus on information-disclosure intention.

4.1 | Study 2a

4.1.1 | Design and Participants

Study 2a used a single factor 2 (anthropomorphism: high, low) between-subjects design, in which regulatory focus was measured as a chronic individual characteristic. One hundred ninety-three consumers (58.5% female; $M_{age} = 31.5$ years, SD = 7.21; see Appendix F in Supporting Information) were recruited via Credamo (http://credamo.com) to participate in the experiment in return for a nominal payment. Credamo is a well-recognized online survey crowdsourcing platform in China that is similar to Amazon Mechanical Turk. It offers a 3-million-member panel, rapid data collection, and built-in quality controls. Researchers can run fully randomized factorial experiments with a demographically diverse sample. Studies using Credamo samples have appeared in a number of top-tier marketing journals (Cao et al. 2025; Zhou et al. 2024).

During the experiment, participants were randomly assigned to one of the two conditions (anthropomorphism: high, low). To ensure data quality, we implemented a multi-layer quality-control protocol: (1) only respondents with a Credamo credit score above 0.8 were allowed to participate, indicating a proven record of reliable responding; (2) a 1-km geo-fence ensured that no two participants were located within 1 km of each other, preventing clustering and the use of coordinated bots; (3) two attention-check items were embedded, and failure on either led to automatic exclusion; and (4) any completion taking less than 60 s was discarded.

4.1.2 | Stimuli and Procedure

In Study 2a, chatbot anthropomorphism was manipulated by similar approaches as in Study 1 (see Appendix B in Supporting Information). However, different from Study 1, an anthropomorphic headphone, instead of a female portrait, was used as the profile photo in the high anthropomorphism condition. A pretest ($n = 75$) was performed to confirm if the manipulations were appropriate. Participants of the pretest were asked to rate the perceived anthropomorphism (inter-item correlation: $r = 0.618$, $p < 0.001$), positive affect ($\alpha = 0.877$), and negative affect ($\alpha = 0.780$) after viewing the stimuli. Results showed that the chatbot in the high anthropomorphism condition triggered higher perceived anthropomorphism than that in the low anthropomorphism condition ($M_{high\text{-}anthr} = 4.86$, $M_{low\text{-}anthr} = 3.89$; $F(1, 73) = 7.278$, $p = 0.009$). However, positive and negative affect did not differ between the high and low anthropomorphism conditions (positive affect: $M_{high\text{-}anthr} = 3.46$, $M_{low\text{-}anthr} = 3.30$; $F(1, 73) = 0.502$, $p = 0.481$; negative affect: $M_{high\text{-}anthr} = 1.34$, $M_{low\text{-}anthr} = 1.39$; $F(1, 73) = 0.191$, $p = 0.663$).

In the main experiment, participants were asked to imagine themselves in a scenario in which an e-commerce website they had used before launched an AI-based product recommendation system and invited customers to interact with the chatbot. Then participants witnessed a screenshot that showed the participant's interaction with the chatbot. After reading the materials, participants had to evaluate their information-disclosure intention ($\alpha = 0.904$), perceived benefits ($\alpha = 0.806$), and perceived

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risks ($\alpha = 0.916$). All variables were measured by the same items as reported in Study 1. Next, participants assessed their regulatory focus and provided demographic information. The regulatory focus was measured by eight items adapted from Lockwood et al. (2002) (see Appendix E in Supporting Information). The average score of the items for prevention ($\alpha = 0.776$) and promotion ($\alpha = 0.636$) focus was separately computed. The score for prevention focus was subtracted from that for promotion focus to form a composite index of the regulatory focus of the participants. A higher final score indicated a predominant promotion focus, while a lower score suggested a predominant prevention focus. This practice fully concords with extant literature (Kim et al. 2023; Kordrostami et al. 2021).

4.1.3 | Results

4.1.3.1 | Main Effect of Anthropomorphism. One-way ANOVAs showed that participants reported higher information-disclosure intention ($M_{\text{high-anthr}} = 4.76$, $M_{\text{low-anthr}} = 4.28$; $F(1, 191) = 6.422$, $p = 0.012$) and perceived benefits ($M_{\text{high-anthr}} = 5.15$, $M_{\text{low-anthr}} = 4.82$; $F(1, 191) = 4.323$, $p = 0.039$) in the high (vs. low) anthropomorphism condition, supporting H1 again. However, anthropomorphism did not affect perceived risks ($M_{\text{high-anthr}} = 5.10$, $M_{\text{low-anthr}} = 5.26$; $F(1, 191) = 0.697$, $p = 0.405$).

4.1.3.2 | Moderation Effects. A 95% bias-corrected bootstrapping analysis with 5000 resamples (PROCESS model 1) was conducted to examine the moderating effect of regulatory focus. The results showed that anthropomorphism and regulatory focus interactively impacted information-disclosure intention ($b = 0.49$, $p = 0.001$). For prevention-focused consumers (−1SD), the impact of anthropomorphism on privacy disclosure was not significant ($M_{\text{high-anthr}} = 4.25$, $M_{\text{low-anthr}} = 4.37$; $b = -0.11$; 95% CI [−0.6278, 0.4003]); while for promotion-focused consumers (+1SD), the high (vs. low) anthropomorphism condition triggered higher information-disclosure intention ($M_{\text{high-anthr}} = 5.28$, $M_{\text{low-anthr}} = 4.19$; $b = 1.09$; 95% CI [0.5799, 1.6062]), supporting H3. A floodlight analysis using the Johnson–Neyman technique further revealed that, for consumers scoring above 0.21 on the regulatory focus index, high (vs. low) anthropomorphism substantially elevated information-disclosure intention.

4.1.3.3 | Moderated Mediation Effects. A 95% bias-corrected bootstrapping analysis with 5000 resamples (PROCESS model 8) was performed. Perceived benefits and perceived risks were specified as parallel mediators of the anthropomorphism × regulatory focus interaction on information-disclosure intention. The interaction significantly predicted perceived benefits ($b = 0.36$, $p = 0.005$) and, inversely, perceived risks ($b = 0.32$, $p = 0.033$). Perceived benefits exerted a positive effect on information-disclosure intention ($b = 0.78$, $p < 0.001$), whereas perceived risks exerted a negative effect ($b = -0.16$, $p = 0.003$). The conditional indirect effect of the interaction between anthropomorphism and regulatory focus on participants' information-disclosure intention through perceived benefits was significant (index of moderated mediation = 0.28, 95% CI [0.0230, 0.5570]). Decomposing this effect, perceived benefits was a significant mediator for promotion-focused participants ($b = 0.61$, 95% CI [0.1612, 1.0672]) but not for prevention-focused participants ($b = -0.08$, 95% CI [−0.4681, 0.2744]). In contrast, the indirect pathway through perceived risks was nonsignificant ($b = 0.05$, 95% CI [−0.0060, 0.1403]).

4.1.4 | Discussion

In Study 2a, we replicated the finding in Study 1 that individuals exhibited higher information-disclosure intention when the chatbot anthropomorphism was high (vs. low). In addition, Study 2a provided evidence for the proposition that regulatory focus moderated this relationship. To enhance the external validity and practical relevance of the findings, Study 2b manipulates regulatory focus situationally and re-examines the underlying mechanisms.

4.2 | Study 2b

4.2.1 | Design and Participants

A total of 206 consumers (65.5% female; $M_{\text{age}} = 30.4$ years, SD = 7.28; see Appendix F in Supporting Information) from Credamo completed the experiment in return for a nominal payment. They were randomly assigned to one of the four conditions in a 2 (anthropomorphism: high, low) × 2 (regulatory focus: prevention-focused, promotion-focused) between-subjects design, in which regulatory focus was manipulated as a situational trait. The quality-control protocol applied in Study 2a was retained.

4.2.2 | Stimuli and Procedure

Similar to the first study, chatbot anthropomorphism was manipulated by chatbot profile photo, name, first or third-person description, and conversational style (see Appendix C in Supporting Information). A pretest ($n = 67$) was performed to confirm if the manipulation was appropriate. Participants of the pretest were asked to rate the perceived anthropomorphism (inter-item correlation: $r = 0.645$, $p < 0.001$), positive affect ($\alpha = 0.854$), and negative affect ($\alpha = 0.652$) after viewing the stimuli. Results showed that the chatbot in the high anthropomorphism condition triggered higher perceived anthropomorphism than that in the low anthropomorphism condition ($M_{\text{high-anthr}} = 5.56$, $M_{\text{low-anthr}} = 4.06$; $F(1, 65) = 17.369$, $p < 0.001$). However, positive and negative affect did not differ between the high and low anthropomorphism conditions (positive affect: $M_{\text{high-anthr}} = 3.71$, $M_{\text{low-anthr}} = 3.33$; $F(1, 65) = 2.473$, $p = 0.121$; negative affect: $M_{\text{high-anthr}} = 1.37$, $M_{\text{low-anthr}} = 1.20$; $F(1, 65) = 2.389$, $p = 0.127$).

During the formal experiment, a dual-task priming approach commonly used in prior research was adopted to evoke situational regulatory focus (Hou et al. 2024; Li et al. 2022; Zhu et al. 2020). This dual task consisted of two components: a recall task and a maze game, each of which targets the core motivational mechanisms of regulatory focus theory (Higgins et al. 1997). While each task has been independently validated for inducing situational regulatory focus (Friedman and Förster 2001; Kim et al. 2023), the dual-task paradigm (Li

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et al. 2022; Zhu et al. 2020) was selected over single-task designs for two key reasons. First, it reinforces the activation of participants' regulatory focus, making the induced promotion or prevention orientation more pronounced and durable compared to a single-task design. Second, it reduces the probability of manipulation ineffectiveness that could stem from the inherent weaknesses of any individual task, thus enhancing the reliability of our regulatory focus manipulation. Specifically, in the recall task, participants in the prevention-focused group were instructed to recall and write down three duties or obligations that were important to them, while those in the promotion-focused group were instructed to list three hopes or dreams. The literature suggests these two types of goals elicit distinct regulatory orientations: pursuit of oughts (e.g., duties) engenders a prevention focus characterized by avoiding negative outcomes, whereas pursuit of ideals (e.g., hopes) fosters a promotion focus manifested in approaching positive outcomes (Higgins et al. 1994; Pham and Avnet 2004). Afterward, participants had to complete a maze game (see Appendix D in Supporting Information): for the prevention-focused group, the task was to guide a mouse to escape from an owl, which activates the security-seeking schema; whereas for the promotion-focused group, the task was to seek out a route for a mouse to get the cheese, which primes the schema of nurturance-seeking and fulfillment of desires (Friedman and Förster 2001).

After completing the dual task, participants had to respond to two items used as a manipulation check of regulatory focus (prevention focus: "Currently, the thoughts and feelings that come to my mind are reflective of duties and responsibilities"; promotion focus: "At present, the thoughts and feelings that come to my mind are reflective of hopes and aspirations" from Ashraf et al. 2016).

Subsequently, participants were told to complete a seemingly unrelated task. They had to imagine themselves in a scenario in which an e-commerce website they had used before launched an AI-based product recommendation system (chatbot). Then, participants witnessed a screenshot that showed the participant's interaction with the chatbot and answered questions regarding information-disclosure intention ($\alpha = 0.884$), perceived benefits ($\alpha = 0.706$), and perceived risks ($\alpha = 0.949$). Finally, participants provided demographic information.

4.2.3 | Results

4.2.3.1 | Manipulation Check. Consistent with Study 2a, we obtained a composite index of regulatory focus by subtracting the score of prevention focus from that of promotion focus. An independent sample *t*-test indicated that the composite index of the promotion-focused group was higher than that of the prevention-focused group ($M_{\text{prevention}} = -2.42$, $M_{\text{promotion}} = 2.04$; $F(1, 204) = 183.263$, $p < 0.001$). Thus, the manipulation of regulatory focus was effective.

4.2.3.2 | Main Effects and Moderation Effects. *Information-disclosure intention*: Anthropomorphism had a significant main effect on information-disclosure intention ($M_{\text{high-anthr}} = 5.62$, $M_{\text{low-anthr}} = 5.10$; $F(1, 202) = 10.657$, $p = 0.001$). Participants reported greater information-disclosure intention when the anthropomorphism was high (vs. low), supporting H1 again. We also observed a significant interaction effect between anthropomorphism and regulatory focus ($F(1, 202) = 6.935$, $p = 0.009$). Simple contrasts revealed that for prevention-focused consumers, the influence of anthropomorphism on information-disclosure intention was not significant ($M_{\text{high-anthr}} = 5.44$, $M_{\text{low-anthr}} = 5.34$; $p = 0.661$); for promotion-focused consumers, the high (vs. low) anthropomorphism condition led to higher information-disclosure intention ($M_{\text{high-anthr}} = 5.77$, $M_{\text{low-anthr}} = 4.84$; $p < 0.001$; see Figure 3), supporting H3 again.

Perceived benefits: The main effect of anthropomorphism on perceived benefits was significant ($M_{\text{high-anthr}} = 5.95$, $M_{\text{low-anthr}} = 5.57$; $F(1, 202) = 11.254$, $p = 0.001$). The interaction between anthropomorphism and regulatory focus was also significant ($F(1, 202) = 4.227$, $p = 0.041$). Specifically, for prevention-focused consumers, the impact of anthropomorphism on perceived benefits was not significant ($M_{\text{high-anthr}} = 5.82$, $M_{\text{low-anthr}} = 5.67$; $p = 0.366$); while for promotion-focused consumers, the high (vs. low) anthropomorphism condition induced higher perceived benefits ($M_{\text{high-anthr}} = 6.06$, $M_{\text{low-anthr}} = 5.47$; $p < 0.001$).

Perceived risks: Chatbot anthropomorphism significantly affects perceived risks ($M_{\text{high-anthr}} = 3.41$, $M_{\text{low-anthr}} = 4.01$; $F(1, 202) = 6.422$, $p = 0.012$). But the interaction effect between anthropomorphism and regulatory focus on perceived risks was not significant ($F(1, 202) = 0.014$, $p = 0.907$).

4.2.3.3 | Moderated Mediation Effects. We performed a bootstrapping analysis using PROCESS model 8 (5000 resamples) to test if perceived benefits and perceived risks mediated the interaction of anthropomorphism and regulatory focus on information-disclosure intention. Perceived benefits and perceived risks were entered into the model as joint mediators. As Figure 3 shows, the interaction between anthropomorphism and regulatory focus predicted perceived benefits ($b = 0.45$, $p = 0.041$) but not perceived risks ($b = -0.05$, $p = 0.907$), and perceived benefits positively influenced information-disclosure intention ($b = 0.90$, $p < 0.001$). The conditional indirect effect of the interaction between anthropomorphism and regulatory focus on participants' information-disclosure intention through perceived benefits was significant (index of moderated mediation = 0.41, 95% CI [0.0096, 0.8211]): for prevention-focused consumers, the mediating effect of perceived benefits was not significant ($b = 0.13$, 95% CI [−0.1753, 0.4161]); for promotion-focused consumers, the mediating effect of perceived benefits was significant ($b = 0.53$, 95% CI [0.2724, 0.8209]). However, the indirect path through perceived risks was not significant ($b = 0.01$, 95% CI [−0.1871, 0.2088]).

4.2.4 | Discussion

In Study 2b, we replicated the main findings of Study 2a by manipulating regulatory focus as a situational trait. By increasing the overall sample size of the study, adopting different experimental materials, and operationalizing regulatory focus in a different approach, Study 2b enhanced the external validity (i.e., the generalizability) of the findings. The findings also

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underscore the practical utility of situational regulatory focus primes in shaping consumers' responses to anthropomorphic chatbots.

5 | General Discussion

This study explored the effect of recommendation chatbot anthropomorphism on consumers' information-disclosure intention, the underlying psychological mechanism, and the moderating role of regulatory focus. Results from three studies offered convergent evidence that chatbot anthropomorphism had a positive influence on consumers' information disclosure intention. Study 1 revealed that the positive effect of chatbot anthropomorphism on consumers' information-disclosure intention is mediated by enhanced perceived benefits. Studies 2a and 2b indicated that the impact of chatbot anthropomorphism on consumers' information-disclosure intention varies with consumers' regulatory focus: chatbot anthropomorphism positively affects information-disclosure intention for promotion-focused consumers, whereas prevention-focused consumers remain unaffected. Additionally, the results of Studies 2a and 2b confirmed that the interactive effect of anthropomorphism and regulatory focus on intention operates solely via perceived benefits.

However, chatbot anthropomorphism did not exert any influence on information-disclosure intention through perceived risks. Likewise, in the moderated-mediation analyses, the interaction between anthropomorphism and regulatory focus did not affect disclosure intention via perceived risks. The null indirect effect through perceived risk may be attributed to two potential mechanisms. First, consumers appear to have become somewhat desensitized to the risks of disclosing information to AI. Even when a chatbot looks or speaks like a human, users may still believe that its data handling remains "non-human" and impersonal; consequently, anthropomorphism does little to alter their baseline privacy risks. Second, prior work offers conflicting predictions: some studies show that anthropomorphism increases perceived control and therefore lowers risk (Chan and Gohary 2022; Kim and McGill 2011), while others suggest that highly human-like agents are seen as autonomous and capable of covertly storing and using personal data, thereby heightening risk (Gkinko and Elbanna 2022). These opposing forces may cancel each other out, leaving no net effect on perceived risk.

5.1 | Theoretical Contributions

This study has several important contributions to the literature. First, it contributes to the literature on privacy disclosure because this study sheds new light on the role of AI-based chatbot anthropomorphism in promoting consumers' intention to disclose personal information. It is widely recognized by researchers that consumers' information disclosure is the key for AI-based agents to provide customized conversations and reliable recommendations (Al-Natour et al. 2021; Kim et al. 2022). Therefore, identifying previously underexplored factors that might impact consumers' information disclosure is crucial in the AI-based service context. Though a few recent studies have attempted to examine the potential role of anthropomorphism in privacy-related issues, they have not been able to illuminate the relationship between anthropomorphism and disclosure as well as the underlying mechanisms (Aw et al. 2024; Pizzi et al. 2023; Uysal et al. 2022; Xie and Lei 2022). This study adds to the expanding body of research on this timely and prominent issue by unraveling the positive effect of chatbot anthropomorphism on consumers' information-disclosure intention through the mediating role of perceived benefits.

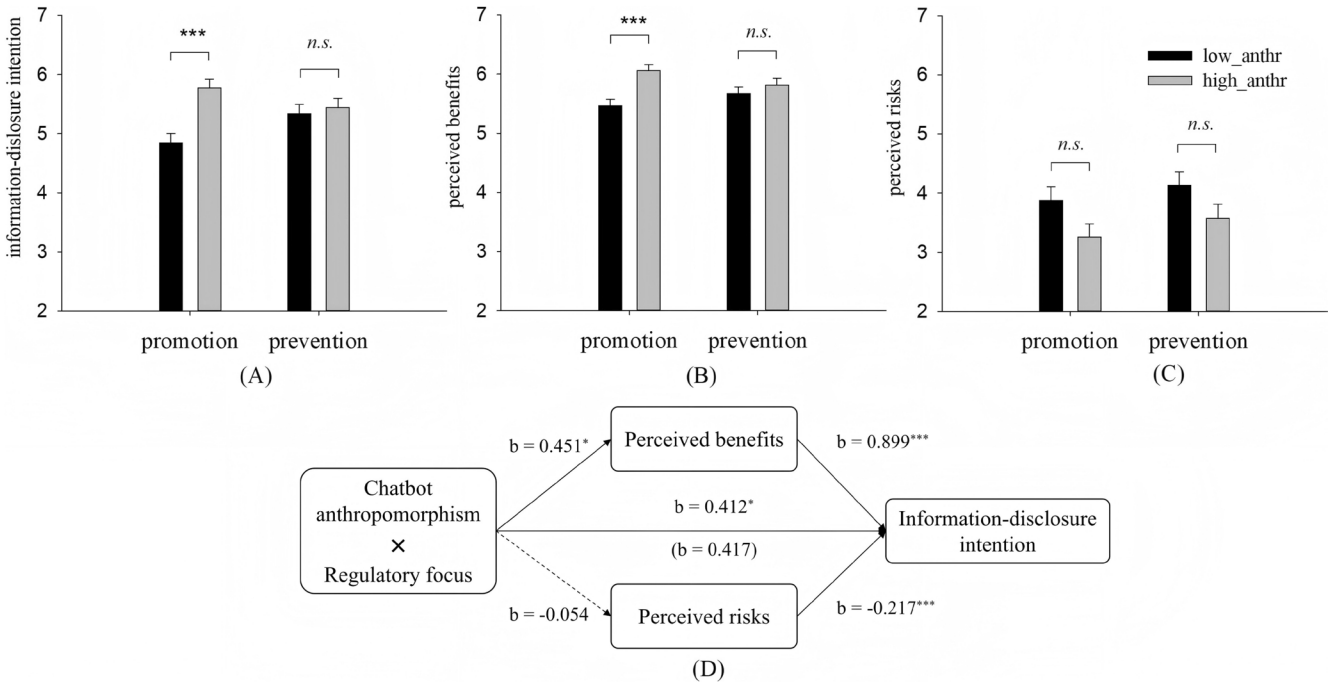


FIGURE 3 | Results of Study 2b. (A) Information-disclosure intention; (B) perceived benefits; (C) perceived risks. The error bars denote standard error of the mean. (D) Moderated mediation results. The value in parentheses indicates the aggregated mediation effect. *$p < 0.05$, ***$p < 0.001$.

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Second, we advance the anthropomorphism literature by enriching our understanding of the downstream effect of anthropomorphism. Anthropomorphism of AI-based agents has been extensively studied regarding its impact on consumers' psychological perceptions (e.g., social presence, trust, and perceived usefulness) (Li and Wang 2022; Munnukka et al. 2022; Yang et al. 2021), behavioral intentions (e.g., purchase intention and usage intention) (Jin and Youn 2023; Konya-Baumbach et al. 2023; Qiu and Benbasat 2009), and actual behaviors (e.g., transaction conversion) (Schanke et al. 2021). However, little is known about the downstream effect of chatbot anthropomorphism on consumers' information-disclosure intention. The findings of the present study bridge the gap in the literature and broaden the scope of anthropomorphism research.

Third, by unraveling the moderating role of regulatory focus in the anthropomorphism—information disclosure association, this study extends the stream of work demonstrating that regulatory focus influences benefits and risks-related decisions. Prior research has investigated how regulatory focus affects consumption decisions using varied research contexts and factors of interest (Khan et al. 2024; Kim et al. 2023; Wang et al. 2020; Zhou and Pham 2004). We link regulatory focus theory to the privacy calculus theory and suggest that regulatory focus moderates the relationship between chatbot anthropomorphism and consumers' information-disclosure intention. Promotion-focused consumers are more likely to perceive the benefits they could gain from sharing personal information with a highly (vs. less) anthropomorphized chatbot and thus show greater information-disclosure intention. By contrast, prevention-focused consumers show similar levels of perceived benefits, perceived risks, and information-disclosure intention when interacting with chatbots with different degrees of anthropomorphism.

5.2 | Practical Implications

The findings of the current study bear meaningful implications for managerial practice. First, given the critical role of consumers' personal data in producing reliable product and service recommendations, marketers should consider anthropomorphizing AI-based chatbots to enhance consumers' intention to disclose personal information. AI-based chatbots are increasingly applied by marketers to deliver high-quality consumer service, facilitate consumer engagement, and gain competitive advantages. This study indicates that chatbot anthropomorphism elevates consumers' information disclosure intention, which enables more tailored and relevant recommendations. Therefore, anthropomorphic features such as human-like appearances and capabilities could be assigned to AI-based chatbots.

Second, marketers can leverage regulatory focus as a strategic managerial input to maximize the effectiveness of chatbot anthropomorphism. Our findings imply that the impact of chatbot anthropomorphism on consumers' information-disclosure intention is contingent upon consumers' regulatory focus: it is effective for promotion-focused consumers, but not for prevention-focused ones. Practitioners should therefore identify users' regulatory orientation and preferentially engage promotion-focused individuals with anthropomorphic chatbots. More importantly, as regulatory focus can be situationally controlled by marketers, they could employ priming strategies to trigger promotion focus prior to chatbot interactions. For example, cues highlighting dreams, aspirations, and accomplishments can be presented to consumers before they are directed to the page where the chatbot provides customized recommendations. Firms might embed a brief recall task within customer-profiling surveys (e.g., asking "What are the three wishes or aspirations you most want to achieve?") with incentives like "Share your dreams—let us help you discover products that bring them to life!" Follow-up activities like a time-sensitive maze game (e.g., "Escape quickly to win a coupon") can further reinforce promotion focus. Notably, while this study used a dual-task priming method, well-designed single-task interventions may yield comparable outcomes. Furthermore, practitioners should align chatbot design with promotion-related features to resonate with promotion-focused users. Specifically, linguistic design, a key aspect of chatbot development, can integrate promotion-focused framing (e.g., using phrases like "I can help you find products that turn your travel dreams into reality!") into anthropomorphic chatbot dialog, which reinforces the connection between information disclosure and benefit attainment.

Third, the mediating role of perceived benefits gives marketers additional insights into promoting consumers' information disclosure when they launch AI-based recommendation systems. According to the present study, perceived benefits, not perceived risks, mediate the relationship between chatbot anthropomorphism and information-disclosure intention. Thus, perceived benefits serve as a key driver of information disclosure. Managers should emphasize value-oriented messages such as "personalized service" and "we understand you," rather than overemphasizing privacy and security, particularly when targeting promotion-focused consumers. Beyond anthropomorphism, marketers can develop other strategies to amplify perceived benefits and encourage disclosure.

5.3 | Limitations and Directions for Future Research

As in any research endeavor, this study has several limitations that suggest opportunities for future studies. First, hypothetical scenarios were used to create the illusory experience of interacting with chatbots. Consumers' perceptions and behaviors might differ when they personally experience the scenarios in the real world. Thus, actual chatbots can be used in future studies to establish more realistic contexts and generalize the findings of the present study.

Second, participants were asked to report their information-disclosure intention without actually disclosing any of their personal data to the chatbot. Further study could be conducted to test if consumers' information-disclosure intention translates into actual disclosure behavior.

Third, although the dual-task paradigm (combining a recall task and a maze game) strengthened the priming of situational regulatory focus and reduced the risk of manipulation failure, it simultaneously obscures the unique contribution of each task. Researchers and marketers cannot determine whether the incremental cost of implementing both tasks is justified, or whether a single, more parsimonious intervention would

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achieve comparable downstream effects. Hence, future research should prioritize single-task paradigms, resorting to dual-task paradigms only when the former proves insufficient.

Fourth, the lack of a significant mediating effect through perceived risk is likely due to multiple interrelated factors. Further research is needed to deconstruct the multidimensional aspects of perceived risk and its antecedents, enabling a more nuanced understanding of its role in human–AI interactions.

Finally, all studies were conducted in China, so generalizability to Western contexts requires consideration. This research's core findings, which are rooted in universally validated theories (i.e., privacy calculus theory and regulatory focus theory) and global similarities in consumer–AI interaction contexts, hold tentative cross-cultural potential. However, cultural differences warrant caution. For instance, consumers' willingness to share personal information is directly shaped by national culture (e.g., power distance and long-term orientation), which also serves as a moderator for the effect of perceived benefits on such willingness (Schumacher et al. 2023). Hence, future research is recommended to replicate this work in Western markets using locally relevant chatbots and test cultural moderators.

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Disclosure

No AI tools were used in the conduct of this research, and no AI tools were used in the process of preparing the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** Supporting Information.

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